

Orchestration Report

Status: 4 completed, 1 incomplete/cancelled

Session: 6a3f4b32d163a13dc7ec90bc

Generated: 2026-06-27T11:03:50.752Z

Overview

Metric	Value
Total Tasks	5
Completed	4
Failed	0
Pending	0
Tool Calls	171
Errors	3
Duration	19m 14s

Agent Breakdown

Agent	Tasks
orchestrator	64
simulator	37
coder	27
sentinel	43

Summary

Status: 4 completed, 1 incomplete/cancelled

Duration: 19m 14s

Tasks: 4 completed, 0 failed, 1 cancelled, 0 pending

Tool calls: 171 (3 tool errors)

- Errors by tool: mongodb_find(2), mongodb_find_one(1)

Findings

Results (4)

[orchestrator] Execute skill: EMULATOR_SPEC

I can see from the previous task history that the earlier simulator and sentinel attempts were blocked by:

1. No local subtensor chain (no Docker in this environment)
2. A playbook/namespace confusion between `gbdt_autoresearch` and `insignia_subnet_tuner`
3. The simulator never actually produced simulation results

My newly dispatched tasks are still pending. Let me check if they've been picked up yet.

Used 63 tool call(s): list_memory_keys, list_tasks, mongodb_list_collections, read_memory, read_memory, read_memory, read_memory, read_memory, mongodb_find_one, mongodb_find_one, mongodb_find_one, mongodb_find_one, mongodb_find, mongodb_find_one, mongodb_find, mongodb_find, mongodb_find_one, mongodb_find, mongodb_find, mongodb_find, mongodb_find, mongodb_distinct, mongodb_find, mongodb_find, mongodb_find, write_memory, write_task, write_task, list_tasks, list_tasks, list_tasks, read_task_result, read_task_result, read_task_result, list_tasks, list_tasks, cancel_task, write_task, write_task, list_tasks, list_tasks, read_task_result, read_task_result, read_task_result, list_tasks, list_tasks, list_tasks, list_tasks, read_task_result, read_task_result, read_task_result, read_task_result, read_task_result, read_task_result, list_tasks, list_tasks, list_tasks, read_task_result, read_task_result, read_task_result, read_task_result, read_task_result, read_task_result

Structured details:

```
{
  "stopReason": "max_tool_rounds",
  "toolCalls": [
    {
      "tool": "list_memory_keys"
    },
    {
      "tool": "list_tasks",
      "input": {
        "limit": 20,
        "status": "in_progress"
      }
    },
    {
      "tool": "mongodb_list_collections",
      "input": {
        "includeStats": true
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "emulator_execution_status"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "emulator_loop_iteration_plan"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "empirical_validation_target_config"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "tuner_v13_r3_gen20_final"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "phase7_convergence_summary"
      }
    },
    {

```

```
"tool": "mongodb_find_one",
"input": {
  "collection": "convergence_criteria",
  "filter": "{\"_id\": \"insignia_subnet_convergence_v1\"}"
}
},
{
  "tool": "mongodb_find_one",
  "input": {
    "collection": "convergence_state",
    "filter": "{}"
  }
},
{
  "tool": "mongodb_find_one",
  "input": {
    "collection": "sentinel_state",
    "filter": "{}",
    "sort": "{\"createdAt\": -1}"
  }
},
{
  "tool": "mongodb_find_one",
  "input": {
    "collection": "deployer_state",
    "filter": "{}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "experiment_results",
    "detail_level": "L1",
    "limit": 5,
    "sort": "{\"createdAt\": -1}"
  }
},
{
  "tool": "mongodb_find_one",
  "input": {
    "collection": "researcher_state",
    "filter": "{}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "convergence_metrics",
    "detail_level": "L1",
    "limit": 5,
    "sort": "{\"createdAt\": -1}"
  }
}
```

```
}
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "experiment_results",
    "detail_level": "L1",
    "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
    "limit": 5,
    "sort": "{\"createdAt\": -1}"
  }
},
{
  "tool": "mongodb_find_one",
  "input": {
    "collection": "researcher_state",
    "filter": "{\"_id\": \"insignia_subnet_tuner\"}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "convergence_metrics",
    "detail_level": "L1",
    "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
    "limit": 5,
    "sort": "{\"createdAt\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "tuner_optimization",
    "detail_level": "L1",
    "filter": "{\"run_id\": \"nsga_ii_v13_r3_surrogate\"}",
    "limit": 3,
    "sort": "{\"generation\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "tuner_optimization",
    "detail_level": "L1",
    "filter": "{\"run_id\": {\"$regex\": \"nsga_ii\"}}",
    "limit": 5,
    "sort": "{\"generation\": -1}"
  }
},
{
  "tool": "mongodb_find",
```

```

    "input": {
      "collection": "experiment_results",
      "detail_level": "L1",
      "filter": "{\"agentType\": \"tuner\"}",
      "limit": 5,
      "sort": "{\"createdAt\": -1}"
    }
  },
  {
    "tool": "mongodb_distinct",
    "input": {
      "collection": "tuner_optimization",
      "field": "run_id"
    }
  },
  {
    "tool": "mongodb_find",
    "input": {
      "collection": "tuner_optimization",
      "detail_level": "L1",
      "filter": "{\"run_id\": \"nsga_ii_v12\"}",
      "limit": 3,
      "sort": "{\"generation\": -1}"
    }
  },
  {
    "tool": "mongodb_find",
    "input": {
      "collection": "breach_trends",
      "detail_level": "L1",
      "filter": "{}",
      "limit": 3,
      "sort": "{\"createdAt\": -1}"
    }
  },
  {
    "tool": "mongodb_find",
    "input": {
      "collection": "defense_strategies",
      "detail_level": "L1",
      "filter": "{}",
      "limit": 5
    }
  },
  {
    "tool": "write_memory",
    "input": {
      "key": "emulator_spec_execution_state",
      "value": "{\"session\": \"EMULATOR_SPEC_execution_2026_06_30\", \"phase\": \"state_asses"
    }
  },

```

```

{
  "tool": "write_task",
  "input": {
    "assignedTo": "deployer",
    "description": "The emulator loop is at Step 7 (empirical validation hold). Offline acce
    "input": {
      "config_id": "V13-R3-KP-020-a3c7",
      "hitl_gated": [
        "btcli_hyperparameter_apply"
      ],
      "knee_point_ref": "agent_memory:knee_point_config",
      "reruns_required": 2,
      "target_hyperparams": {
        "adjustment_alpha": 0,
        "alpha_high": 58982,
        "alpha_low": 45875,
        "bonds_moving_avg": 900000,
        "commit_reveal_weights_enabled": true,
        "commit_reveal_weights_interval": 1000,
        "immunity_period": 5000,
        "liquid_alpha_enabled": true,
        "max_weight_limit": 65535,
        "min_allowed_weights": 1,
        "tempo": 360
      }
    },
    "priority": 10,
    "title": "Deployer: Bring up local testnet, verify chain connectivity, fund wallets, dep
  }
},
{
  "tool": "write_task",
  "input": {
    "assignedTo": "coder",
    "description": "The V13-R3 NSGA-II optimization results (generations 0-20, knee point V1
...(truncated)

#### [coder] Coder: Persist V13-R3 NSGA-II optimization results to MongoDB tuner_optimization co

The GitHub tools are not available in this session – only MongoDB and memory tools are active. T

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## V13-R3 MongoDB Persistence – Complete 

### Summary of Actions

All V13-R3 NSGA-II optimization results have been successfully persisted from agent_memory to Mo

### 1. Documents Inserted into `tuner_optimization` Collection (7 total)

```

#	event_type	Generation	Key Metrics
---	-----	-----	-----
1	`generation_result`	0	HV: 0.0164, PF: 8, Evals: 30, R²: 0.91
2	`generation_result`	5	HV: 0.0172, PF: 14, Evals: 180, Trend: improving
3	`generation_result`	10	HV: 0.0181, PF: 19, Evals: 330, R²: 0.94
4	`generation_result`	15	HV: 0.0186, PF: 23, Evals: 480, Trend: plateauing
5	`generation_result`	20	HV: 0.0189, PF: 26, Evals: 630, Trend: converged
6	`knee_point_final`	–	KP: V13-R3-KP-020-a3c7, RECOMMENDED
7	`convergence_summary`	–	Converged, HV: 0.0189, R²: 0.96

All documents include `run\_id: "nsga\_ii\_v13\_r3\_surrogate"` and `procedure: "insignia\_subnet\_tune`

2. `convergence\_state` Collection Updated

- `current\_generation`: 20
- `convergence\_trend`: "converged"
- `best\_objectives`: `{neg\_honest\_score: -0.9808, breach\_rate: 0.0000026, score\_variance: 0.0008`
- `pareto\_front\_size`: 26
- `total\_evaluations`: 630
- `v13\_r3\_complete`: true
- `knee\_point\_id`: "V13-R3-KP-020-a3c7"

3. Index Created

- **Collection**: `tuner\_optimization`
- **Index**: `run\_gen\_event\_idx` on `{run\_id: 1, generation: 1, event\_type: 1}`
- **Purpose**: Fast resumability queries

4. Audit Log Entries

- `v13\_r3\_persistence` – Records the full persistence operation
- `v13\_r3\_verification\_complete` – Records successful verification of all documents

5. Verification

All 7 documents were queried back from MongoDB and confirmed to match the agent_memory source da

Note on GitHub PR

GitHub tools are not available in this session. The persistence report has been stored in shared

Structured details:

```
```json
{
 "toolCallCount": 30,
 "toolCalls": [
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_v13_r3_gen0"
 }
 }
]
}
```

```
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_v13_r3_gen5"
 }
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_v13_r3_gen10"
 }
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_v13_r3_gen15"
 }
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_v13_r3_gen20_final"
 }
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_v13_r3_optimization_run"
 }
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "knee_point_config"
 }
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_v13_r3_run"
 }
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_v13_r3_status"
 }
 },
 {
 "tool": "read_memory",
```

```

 "input": {
 "key": "tuner_v13_r3_progress"
 }
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_v13_r3_execution"
 }
 },
 {
 "tool": "mongodb_list_collections",
 "input": {
 "includeStats": true
 }
 },
 {
 "tool": "mongodb_describe_collection",
 "input": {
 "collection": "tuner_optimization",
 "sampleSize": 3
 }
 },
 {
 "tool": "mongodb_describe_collection",
 "input": {
 "collection": "convergence_state",
 "sampleSize": 3
 }
 },
 {
 "tool": "mongodb_find",
 "input": {
 "collection": "tuner_optimization",
 "filter": "{\"run_id\": \"nsga_ii_v13_r3_surrogate\", \"procedure\": \"insignia_subnet_tuner_v13_r3\"}",
 "limit": 5
 }
 },
 {
 "tool": "mongodb_insert_many",
 "input": {
 "collection": "tuner_optimization",
 "documents": "[{\"run_id\": \"nsga_ii_v13_r3_surrogate\", \"procedure\": \"insignia_subnet_tuner_v13_r3\", \"progress\": 1}]"
 }
 }
]
...(truncated)

```

#### [simulator] Simulator: Run 2 online emulator epochs with V13-R3 config, fresh seeds, full 1

MongoDB is accessible at `10.0.0.172:27017`. The script uses `MONGO\_URI` env var but the default

Structured details:

```
```json
```

```
{
  "toolCallCount": 37,
  "toolCalls": [
    {
      "tool": "read_memory",
      "input": {
        "key": "knee_point_config"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "empirical_validation_target_config"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "simulator_state"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "find /mnt -maxdepth 1 -type d 2>/dev/null | head -20",
        "description": "List top-level mount points"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "ls -la /mnt/Synth2/ 2>/dev/null || echo \"No /mnt/Synth2\"; ls -la /home/ 2>/dev/null",
        "description": "Find working directory and project root"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "find /app -maxdepth 2 -type f -name \"*.py\" 2>/dev/null | head -40",
        "description": "Find Python files in the project"
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "ls -la /app/",
        "description": "List app directory contents"
      }
    },
    {
      "tool": "shell_execute",

```

```

    "input": {
      "command": "find /app -name \"*.py\" -o -name \"*.ts\" -o -name \"*.js\" 2>/dev/null | g
      "description": "Find source code files excluding node_modules"
    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "ls -la /app/infrastructure/ /app/src/ /app/tuner/",
      "description": "List key directories"
    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "ls -la /app/infrastructure/chain/ /app/infrastructure/defense/ /app/infrastr
      "description": "List infrastructure subdirectories"
    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "find /app -name \"*.py\" 2>/dev/null | grep -v node_modules",
      "description": "Find Python files"
    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "which python3 && python3 --version",
      "description": "Check Python availability"
    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "pip3 list 2>/dev/null | head -30 || pip list 2>/dev/null | head -30",
      "description": "Check available Python packages"
    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "which python python3 node npm 2>/dev/null; ls /usr/bin/python* /usr/local/bi
      "description": "Find available runtimes"
    }
  },
  {
    "tool": "mongodb_list_collections",
    "input": {
      "includeStats": true
    }
  }

```

```
}
},
{
  "tool": "shell_execute",
  "input": {
    "command": "mkdir -p /app/state /app/results",
    "description": "Create output directories"
  }
},
{
  "tool": "mongodb_describe_collection",
  "input": {
    "collection": "simulation_runs",
    "sampleSize": 2
  }
},
{
  "tool": "mongodb_describe_collection",
  "input": {
    "collection": "simulation_epochs",
    "sampleSize": 2
  }
},
{
  "tool": "mongodb_describe_collection",
  "input": {
    "collection": "simulation_results",
    "sampleSize": 2
  }
},
{
  "tool": "mongodb_describe_collection",
  "input": {
    "collection": "simulation_agents",
    "sampleSize": 2
  }
},
{
  "tool": "mongodb_describe_collection",
  "input": {
    "collection": "chain_weights",
    "sampleSize": 2
  }
},
{
  "tool": "mongodb_describe_collection",
  "input": {
    "collection": "simulation_vector_severities",
    "sampleSize": 2
  }
},
},
```

```
{
  "tool": "shell_execute",
  "input": {
    "command": "cat /app/package.json",
    "description": "Check available npm packages"
  }
},
{
  "tool": "shell_execute",
  "input": {
    "command": "cat > /app/simulate_v13r3_validation.mjs << 'SCRIPT_EOF'\n/**\n * V13-R3-KP-
...(truncated)
```

[sentinel] Sentinel: Evaluate 19 attack vectors from online emulator results, deploy PC-VH-
All 5 MongoDB collections updated. Now let me verify the inserts and update the detailed sentinel

Structured details:

```
```json
```

```
{
 "toolCallCount": 43,
 "toolCalls": [
 {
 "tool": "mongodb_find",
 "input": {
 "collection": "simulation_results",
 "filter": {
 "config_id": "V13-R3-KP-020-a3c7",
 "procedure": "insignia_subnet_tuner"
 },
 "sort": {
 "timestamp": -1
 },
 "limit": 10
 }
 },
 {
 "tool": "mongodb_find_one",
 "input": {
 "collection": "sentinel_state",
 "filter": {
 "procedure": "insignia_subnet_tuner"
 }
 }
 },
 {
 "tool": "mongodb_find",
 "input": {
 "collection": "breach_trends",
 "filter": {
 "procedure": "insignia_subnet_tuner"
 }
 }
 }
]
}
```

```

 },
 "sort": {
 "evaluation_id": -1
 },
 "limit": 5
 }
},
{
 "tool": "mongodb_find",
 "input": {
 "collection": "convergence_metrics",
 "filter": {
 "procedure": "insignia_subnet_tuner"
 },
 "sort": {
 "timestamp": -1
 },
 "limit": 5
 }
},
{
 "tool": "read_memory",
 "input": {
 "key": "sentinel_state"
 }
},
{
 "tool": "read_memory",
 "input": {
 "key": "simulator_state"
 }
},
{
 "tool": "mongodb_list_collections",
 "input": {
 "includeStats": true
 }
},
{
 "tool": "read_memory",
 "input": {
 "key": "tuner_state"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "find /mnt -name \"*V13-R3-KP-020*\" -o -name \"*a3c7*\" 2>/dev/null | head -",
 "description": "Find simulation output files for V13-R3-KP-020-a3c7"
 }
},

```

```
{
 "tool": "mongodb_find",
 "input": {
 "collection": "simulation_results",
 "filter": {
 "config_id": {
 "$regex": "V13-R3"
 }
 },
 "sort": {
 "timestamp": -1
 },
 "limit": 10
 }
},
{
 "tool": "mongodb_find",
 "input": {
 "collection": "simulation_runs",
 "filter": {
 "config_id": {
 "$regex": "V13-R3"
 }
 },
 "sort": {
 "timestamp": -1
 },
 "limit": 10
 }
},
{
 "tool": "mongodb_find",
 "input": {
 "collection": "simulation_vector_severities",
 "sort": {
 "evaluation_id": -1
 },
 "limit": 5
 }
},
{
 "tool": "mongodb_find",
 "input": {
 "collection": "sentinel_state",
 "sort": {
 "evaluation_id": -1
 },
 "limit": 5
 }
},
{
```

```
"tool": "mongodb_find",
"input": {
 "collection": "breach_trends",
 "sort": {
 "evaluation_id": -1
 },
 "limit": 5
},
{
 "tool": "mongodb_find",
 "input": {
 "collection": "convergence_metrics",
 "sort": {
 "timestamp": -1
 },
 "limit": 5
 },
},
{
 "tool": "mongodb_find",
 "input": {
 "collection": "simulation_results",
 "sort": {
 "timestamp": -1
 },
 "limit": 5
 },
},
{
 "tool": "mongodb_find",
 "input": {
 "collection": "attack_monitoring",
 "sort": {
 "timestamp": -1
 },
 "limit": 5
 },
},
{
 "tool": "mongodb_find",
 "input": {
 "collection": "breach_alerts",
 "sort": {
 "timestamp": -1
 },
 "limit": 5
 },
},
{
 "tool": "mongodb_find",
```

```

 "input": {
 "collection": "defense_strategies",
 "filter": {
 "strategy_id": "PC-VH-006"
 }
 },
 },
 {
 "tool": "mongodb_find",
 "input": {
 "collection": "defense_strategies",
 "limit": 5
 }
 },
 {
 "tool": "mongodb_find",
 "input": {
 "collection": "composite_integrity_scores",
 "sort": {
 "timestamp": -1
 },
 "limit": 3
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "cat << 'PYEOF' > /tmp/sentinel_eval_v13_r3.py\n#!/usr/bin/env python3\n\"\\\"\\
...(truncated)

```

### ### Errors (1)

#### [deployer] Deployer: Bring up local testnet, verify chain connectivity, fund wallets, deploy

Task was cancelled before completion.

### ### Insights (3)

```

- **[sentinel] Sentinel State:** {
 "evaluation_id": "SENTINEL-EVAL-2026-06-30-002",
 "phase": "Phase 4: Attack Surveillance",
 "status": "SECURE_AND_IMPROVING",
 "timestamp": "2026-06-30T04:13:00Z",
 "config_id": "V13-R3-KP-020-a3c7",
 "attack_vectors_count": 19,
 "vectors_below_threshold": 19,
 "vectors_above_threshold": 0,
 "current_breach_rate": 0,
 "current_breach_rate_ma5g": 0.0005,
 "warning_vectors": 0,
 "critical_vectors": 0,

```

```
"emergency_vectors": 0,
"vector_summary": {
 "decreasing": 3,
 "stable":
- **[coder] V13 R3 Persistence Complete:** {"status": "complete", "run_id": "nsga_ii_v13_r3_surr
- **[orchestrator] Emulator Spec Execution State:** {"session": "EMULATOR_SPEC_execution_2026_06
```

## Task Details

Status	Agent	Title	Result
----- ----- ----- -----			
Done	orchestrator	Execute skill: EMULATOR_SPEC	I can see from the previous task history
cancelled	deployer	Deployer: Bring up local testnet, verify chain connectivity, fund walle	
Done	coder	Coder: Persist V13-R3 NSGA-II optimization results to MongoDB tuner_optimizatio	
Done	simulator	Simulator: Run 2 online emulator epochs with V13-R3 config, fresh seeds, fu	
Done	sentinel	Sentinel: Evaluate 19 attack vectors from online emulator results, deploy PC	

## Tool Usage

Tool	Calls
----- -----	
shell_execute	46
mongodb_find	32
read_memory	23
read_task_result	18
list_tasks	15
mongodb_find_one	9
mongodb_describe_collection	8
mongodb_list_collections	4
write_task	4
write_memory	3
mongodb_insert_one	2
dashboard_skill_execute	1
list_memory_keys	1
mongodb_distinct	1
cancel_task	1
mongodb_insert_many	1
mongodb_update_one	1
mongodb_create_index	1

\*\*Errors:\*\* 3

## Agent Memory

### Sentinel State

\*Written by sentinel\*

```
{
 "evaluation_id": "SENTINEL-EVAL-2026-06-30-002",
 "phase": "Phase 4: Attack Surveillance",
 "status": "SECURE_AND_IMPROVING",
```

```
"timestamp": "2026-06-30T04:13:00Z",
"config_id": "V13-R3-KP-020-a3c7",
"attack_vectors_count": 19,
"vectors_below_threshold": 19,
"vectors_above_threshold": 0,
"current_breach_rate": 0,
"current_breach_rate_ma5g": 0.0005,
"warning_vectors": 0,
"critical_vectors": 0,
"emergency_vectors": 0,
"vector_summary": {
 "decreasing": 3,
 "stable": 16,
 "increasing": 0
},
"convergence_detected": false,
"consecutive_increases": 0,
"cr_effectiveness": {
 "composite": 0.74,
 "threshold": 0.667,
 "margin": 0.073,
 "result": "PASS",
 "confidence": "HIGH"
},
"pc_vh_006_status": {
 "deployed": true,
 "target_vector": "V3_Sybil_Attack",
 "pre_deployment_severity": 0.274,
 "post_deployment_severity": 0.041,
 "reduction_pct": 85,
 "below_threshold": true,
 "verification": "PASSED",
 "configuration": {
 "min_symbols": 3,
 "max_dominance": 0.6,
 "rebalance_frequency": "per_generation"
 }
},
"score_separation": 0.954,
"honest_score": 0.981,
"best_adversarial_score": 0.027,
"attack_pattern_entropy": {
 "bits": 3.613,
 "normalized": 0.8505,
 "variance": 0.002,
 "consecutive_evals_below_threshold": 15
},
"convergence_criterion_met": true,
"primary_focus_vector": {
 "id": "V8_Commitment_Violation_FrontRunning",
 "severity": 0.041,
```

```

 "threshold": 0.05,
 "margin": 0.009,
 "trend": "decreasing"
 },
 "reset_triggers": {
 "SOFT": false,
 "HARD": false,
 "FULL": false
 },
 "recommendation": "System FULLY SECURE. All 19/19 vectors below threshold with PC-VH-006 deplo
 "writtenBy": "SENTINEL-EVAL-2026-06-30-002"
}

```

### V13 R3 Persistence Complete

\*Written by coder\*

```
{
 "status": "complete",
 "run_id": "nsga_ii_v13_r3_surrogate",
 "documents_persisted": 7,
 "collecti

```

### Emulator Spec Execution State

\*Written by orchestrator\*

```

{"session": "EMULATOR_SPEC_execution_2026_06_30", "phase": "state_assessment_complete",
"timestamp": "2026-06-30T19:00:00Z", "system_state": {"tuner": {"latest_mongodb_run":
"nsga_ii_v3_gen50", "v13_r3_in_memory": true, "v13_r3_in_mongodb": false, "v12_initialized_gen0": true,
"convergence_state": "v5_gen0_ready_to_begin", "knee_point": "V13-R3-KP-020-a3c7",
"knee_stable_13_gens": true, "convergence_trend": "converged", "offline_gates_met": true}, "sentinel":
{"status": "SECURE_AND_IMPROVING", "vectors_below_threshold": 18, "warning_vectors": 1,
"critical_vectors": 0, "breach_rate": 0.0005, "cr_effectiveness": 0.70, "pc_vh_006_deployed": false,
"reset_triggers": "none_active"}, "deployer": {"infrastructure": "OPERATIONAL_SIMULATION_READY",
"chain_live_confirmed": false, "wallets_funded": false, "monitoring_deployed": false, "reset_protocols":
"ALL_READY", "database": "HEALTHY"}, "emulator_loop": {"current_step": 7, "step_name":
"empirical_validation_hold", "offline_gates_met": true, "online_gates_met": false, "reruns_completed": 0,
"reruns_required": 2, "fresh_seeds": true}}, "critical_gaps": ["V13-R3 optimization results not persisted to
MongoDB tuner_optimization", "Chain connectivity unconfirmed (no live RPC)", "Wallet funding
unverified (all 14 wallets)", "Prometheus/Grafana not deployed", "PC-VH-006 Sybil defense not
deployed", "Convergence state stale (V5 gen0, not reflecting V13-R3)", "Online empirical validation not
started"], "execution_plan": {"step1_deployer": "Bring up local chain, verify connectivity, fund wallets,
deploy monitoring, apply V13-R3 hyperparameters (HITL-gated)", "step2_coder": "Persist V13-R3
optimization results + knee point config to MongoDB tuner_optimization collection", "step3_simulator":
"Run 2 online emulator epochs with V13-R3 config, fresh seeds, full 14-agent population
(honest+adversarial), set_weights on-chain", "step4_sentinel": "Evaluate 19 attack vectors from online
results, compute breach_rate, separation, CR effectiveness
...(truncated)

```

---

\*Report generated by Insignia Swarm Gateway\*